PYTHON DEVELOPER CASE STUDY
Task 1:

LLM-Based Database Data Discovery System

PROJECT OVERVIEW

This case study aims to develop a system for automatic discovery and classification of data in PostgreSQL databases using LLM (Large Language Model) technologies. The system will be developed using the FastAPI framework and containerized with Docker.

TECHNICAL REQUIREMENTS

Technology Stack

- Backend Framework: FastAPI
- Database: PostgreSQL (target DB)
- LLM Provider: OpenAI Compatible API
- Containerization: Docker & Docker Compose
- Authentication: Basic Authentication

System Architecture

The system will consist of 6 main service endpoints:

1. Authentication Service

- Endpoint: /auth
- Purpose: Basic authentication for system security
- Functionality: User authentication and token management

2. Database Metadata Extraction & Storage Service

- **Endpoint:**
    - POST /db/metadata
- **Purpose:**
    - Connect to PostgreSQL database, extract and store metadata
- **Input:**
    - Database connection details (host, port, database, username, password)

- **Output:**
    - Metadata ID (unique identifier)
    - Table list
    - Column information for each table (with unique column ID)
    - Column data types
- **Functionality:**
    - Stores metadata in system database
    - Assigns unique ID to each column
    - Securely stores connection information

3. Metadata List Service

- **Endpoint:**
    - GET /metadata
- **Purpose:**
    - List stored metadata records
- **Output:**
    - Metadata IDs
    - Database names
    - Creation dates
    - Table counts

4. Metadata Detail Service

- **Endpoint:**
    - GET /metadata/{metadata_id}
- **Purpose:**
    - Retrieve details of specific metadata
- **Input:**
    - Metadata ID

- **Output:**
    - Table list
    - Column details for each table
    - Column IDs
    - Data types

5. Metadata Delete Service

- **Endpoint: DELETE /metadata/{metadata_id}**
- **Purpose: Delete stored metadata**
- **Input:**
    - Metadata ID
- **Output:**
    - Deletion status

6. Data Classification Service

- **Endpoint:**
    - POST /classify
- **Purpose:**
    - Data classification using LLM
- **Input:**
    - Column ID
    - Sample count (default: 10)
- **Output:**
    - Probability distributions from predefined classes
- **Functionality:**
    - Finds metadata and connection info from Column ID
    - Extracts specified number of sample data
    - Performs classification using LLM

PII Classification Categories

1. **Email Address**
    - **Personal and business email addresses**
    - **Examples: john.doe@gmail.com, user@company.com**
2. **Phone Number**
    - **Mobile, landline, and international phone numbers**
    - **Examples: +1-555-123-4567, (555) 123-4567**
3. **Social Security Number (SSN)**
    - **US Social Security Numbers and similar national identifiers**
    - **Examples: 123-45-6789, 987654321**
4. **Credit Card Number**
    - **Payment card numbers (Visa, MasterCard, etc.)**
    - **Examples: 4532-1234-5678-9012, 5555555555554444**
5. **National ID Number**
    - **Government-issued identification numbers**
    - **Examples: Passport numbers, driver's license numbers, national ID cards**
6. **Full Name**
    - **Complete personal names (first, middle, last)**
    - **Examples: John Michael Smith, Maria Garcia Lopez**
7. **First Name**
    - **Personal first names only**
    - **Examples: John, Maria, Ahmed, Sarah**
8. **Last Name / Surname**
    - **Family names or surnames only**
    - **Examples: Smith, Johnson, Garcia, Wilson**
9. **TCKN (Turkish Citizenship Number)**

- **Turkish Republic Citizenship Number (11-digit identifier)**
- **Examples: 12345678901, 98765432109**

10. Home Address

- **Residential addresses including street, city, state, ZIP**
- **Examples: 123 Main St, Anytown, CA 90210**

11. Date of Birth

- **Personal birth dates in various formats**
- **Examples: 1985-03-15, March 15, 1985, 03/15/1985**

12. IP Address

- **Internet Protocol addresses that can identify individuals**
- **Examples: 192.168.1.1, 2001:0db8:85a3:0000:0000:8a2e:0370:7334**

Task 2:

Convert the entire system developed above into a Java project with the help of AI & LLM tools. Explain all steps methods and prompts that you apply.

SUBMISSION REQUIREMENTS:

- All system configuration including system database connections, basic authentication credentials, LLM connection parameters and other required information must be stored in the .env file
- Working Docker setup with docker-compose
- Comprehensive README with setup instructions
- .env.example file with all required variables

Demo DB script Link:
https://kafeinproduct.blob.core.windows.net/public/DemoDatabaseScript.code.sql